

Prachi Gupta, PhD

Cancer Biology · Immuno-Oncology · Tumor Microenvironment · Translational Cancer Research

PROFESSIONAL SUMMARY

PhD-trained cancer biologist with 7+ years of postdoctoral research experience spanning immuno-oncology, tumor microenvironment biology, extracellular vesicles, T-cell responses, molecular signaling, biomarker discovery, and translational cancer models. Research experience includes ovarian cancer, renal cell carcinoma, melanoma, and breast cancer, with expertise in in vitro and in vivo experimental design, immune-cell functional assays, organoid models, preclinical studies, flow cytometry, molecular biology, and proteomics/transcriptomics analysis.

CORE SCIENTIFIC EXPERTISE

Immuno-Oncology	T-cell and B-cell assays · Cytotoxicity · Immune phenotyping · Primary human TIL culture
Tumor Microenvironment	Tumor-immune interactions · Immune suppression · Extracellular vesicles · S1P/SPHK1 signaling
Preclinical Research	In vitro/in vivo models · Mouse tumor models · ccRCC organoids · Human patient samples
Molecular Biology	Western blot · PCR · Transfection · Immunoprecipitation · ELISA · IHC/IF · Confocal microscopy
Data & Analysis	Multicolor flow cytometry · FlowJo · GSEA · Proteomics · Transcriptomics · Ingenuity Pathway Analysis · Seahorse
Scientific Communication	Experimental design · Protocol development · Grant writing · Manuscripts · Reviews · Scientific presentations

RESEARCH EXPERIENCE

Research Fellow — Earle A. Chiles Research Institute, Portland, Oregon | 2024–Present | PI: Allie Grossmann

Role of ARF6-GTP in tumor-cell intrinsic defense and repair mechanisms to resist T-cell attack.

- Develop and validate experimental methods and protocols investigating tumor-cell responses to immune attack.
- Study cytokine-mediated melanoma cell death and evaluate whether ARF6 knockdown sensitizes melanoma cells to immune attack.
- Maintain melanoma cell lines; organize and present research findings; collaborate across projects and support validation of results.

Research Associate — Memorial Sloan Kettering Cancer Center, New York | 2023–2024 | PI: Emily Cheng

Generating renal carcinoma patient-specific antigenic T cells in vitro using immunogenic cell death.

- Developed a method to generate tumor-reactive CD8 T cells using immunogenic cell death in vitro.
- Developed a clear cell renal cell carcinoma (ccRCC) organoid model and performed microtome sectioning, FFPE processing, and CA9 staining.
- Worked with clinical mouse models of ccRCC to support translational cancer research.

Postdoctoral Fellow — Medical College of Wisconsin, Milwaukee, Wisconsin | 2019–2023 | PI: Sunila Pradeep

Role of SPHK1 in the immunosuppressive tumor microenvironment in ovarian cancer.

- Isolated and characterized extracellular vesicles from ovarian cancer cells and analyzed their proteome.
- Designed and performed in vitro and in vivo experiments addressing tumor progression and immune suppression.
- Investigated S1P-mediated signaling and extracellular-vesicle mechanisms of immune suppression.
- Contributed to grant writing, manuscripts, review articles, and presentations at AACR and institutional scientific meetings.

SELECTED RESEARCH CONTRIBUTIONS

- **Ovarian cancer immunosuppression:** research identified S1P-mediated signaling and extracellular-vesicle mechanisms associated with immune suppression, including effects on PD-L1 expression and CD8 T-cell activity.

- **Patient-relevant models:** experience developing ccRCC organoids, generating tumor-reactive CD8 T cells, and working with clinical mouse models.
- **Melanoma therapy resistance:** current research investigates ARF6-GTP and tumor-cell intrinsic defense and repair mechanisms that may influence resistance to T-cell-mediated attack.

SELECTED PUBLICATIONS

- Gupta P, et al. Tumor Derived Extracellular Vesicles Drive T Cell Exhaustion in Tumor Microenvironment through Sphingosine Mediated Signaling and Impacting Immunotherapy Outcomes in Ovarian Cancer. *Advanced Science*. 2022.
- Mittal S, et al., including Gupta P. eIF4E Enriched Extracellular Vesicles Induce Immunosuppressive Macrophages through HMGR-Mediated Metabolic Rewiring. *Advanced Science*. 2025.
- Wang J, et al., including Gupta P. Rapid activation of ARF6 after RAF inhibition augments BRAFV600E and promotes therapy resistance. *Oncogene*. 2026.
- George J, et al., including Gupta P. RNA-binding protein FXR1 drives cMYC translation by recruiting eIF4F complex to the translation start site. *Cell Reports*. 2021.
- Geethadevi A, et al., including Gupta P. Oncostatin M Receptor–Targeted Antibodies Suppress STAT3 Signaling and Inhibit Ovarian Cancer Growth. *Cancer Research*. 2021.
- Gupta P, et al. B cells as an immune-regulatory signature in ovarian cancer. *Cancers*. 2019.
- Gupta P, et al. Autoantibodies against TYMS and PDLIM1 proteins detected as circulatory signatures in Indian breast cancer patients. *PROTEOMICS–Clinical Applications*. 2016.
- Mittal S, et al., including Gupta P. Protocol for the isolation of tumor cell-derived extracellular vesicles followed by in vivo metastasis assessment in a murine ovarian cancer model. *STAR Protocols*. 2024.

EDUCATION

PhD, Cancer Biology — Council of Scientific and Industrial Research – Indian Institute of Toxicology Research, Lucknow, India | 2010–2018. Dissertation: Characterization of repertoire of humoral immune response and cancer stem cells associated microvesicles in breast cancer.

MS, Biotechnology — Amity University, India | 2007–2010.

BS — CCS University, India | 2004–2007.

AWARDS & RESEARCH FUNDING

- **Ovarian Cancer Research Alliance** — Ann Schreiber Mentored Investigator Award (2021).
- **U.S. Department of Defense, Ovarian Cancer Research Program** — Pilot Award (2021–2023), Key Person on Pradeep (PI) award.

RESEARCH PROFILE

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